



Department of Information Science and Engineering
Acharya Institute of Technology

Acharya Dr. Sarvepalli Radhakrishnan Road, Bengaluru - 560107

MINI PROJECT PROGRESS REPORT

Batch No	26		
Guide	Prof. Vinutha Raghu		
Mini Project Title	NetVisor		
Progress Report No	01		
Date of Submission	21/11/2025		
Date	From: 04/09/2025	To:21/10/2025	

Sl. No.	Students Name	USN	Signature with date
1	Arvind D H	1AY24IS400	
2	Girishkumar N M	1AY24I404	
3	Manoj	1AY24IS406	
4	Premkumar Teli	1AY24IS407	

Progress:

1. About the Project and Progress :

The first stage of the NetVisor project focused on establishing the core architecture and developing the foundational components necessary for real-time network monitoring. The primary objective during this phase was to build a functional prototype capable of detecting active devices on a network, capturing live packet data, and presenting the information through a structured and interactive dashboard.

The system architecture was conceptualized as a modular three-layer model consisting of the Flask-based backend, the web dashboard frontend, and an external packet analysis layer powered by TShark and ARP scanning utilities. Each layer was designed to ensure scalability, modularity, and secure data handling while maintaining seamless communication across components.

Significant progress was made in developing the backend services. Core Flask routes were created to manage session handling, packet capture requests, device scanning operations, and dashboard data retrieval. A REST API structure was also implemented to exchange JSON-based data between the frontend and backend efficiently.

The frontend interface, built using HTML, CSS, Bootstrap 5, and Chart.js, was developed as an intuitive dashboard capable of visualizing real-time traffic statistics, bandwidth usage graphs, and device information tables. The UI also included modules for suspicious activity detection, alerts, and network health status.

One of the major accomplishments in this phase was the successful integration of ARP scanning for device discovery and TShark-based packet capturing for retrieving protocol-level details. This allowed NetVisor to extract real-time information such as IP/MAC addresses, protocol types, packet sources, and communication patterns. A background service module was also implemented to continuously monitor and update traffic data.

By the end of Stage 1, NetVisor achieved a fully functional prototype capable of scanning active devices, capturing packets, and visualizing essential network insights on the dashboard. This milestone established the core monitoring pipeline, laying a strong

foundation for implementing advanced anomaly detection and AI-driven analytics in the subsequent development stages.

2. Problems Faced and Overcomes :

During Stage 1, several challenges emerged involving backend integration, real-time synchronization, and external tool configuration.

The first major issue was related to **TShark integration**. Initial attempts to trigger packet capture externally resulted in permission errors and incomplete output streams. After troubleshooting, the team configured appropriate system-level privileges and streamlined the capture command execution through subprocess handlers, ensuring stable packet retrieval.

Another challenge was ensuring **smooth real-time data updates** on the dashboard. Early tests showed delays and inconsistent refresh cycles, resulting in outdated or partial network statistics. This was resolved by implementing AJAX-based dynamic updates and optimizing API responses for faster polling.

Device discovery through ARP scanning also presented difficulties due to variations in network environments. Some networks required elevated permissions or different scanning strategies. The team improved reliability by validating ARP table entries and adding fallback scanning mechanisms.

Additionally, proper filtering of suspicious traffic was initially inaccurate because raw TShark data required additional parsing. The team refined the parsing logic and introduced pattern-based anomaly detection for better accuracy.

Despite these challenges, each issue strengthened the system's architecture and resulted in a more robust and stable Stage 1 build.

Work schedule and final results

Week 1–2 (September 04 – September 17): Project setup was completed with Flask backend initialization and MySQL database configuration. Basic routing structure and folder organization were established. Testing environment for packet capture was prepared and TShark permissions were configured. Initial system architecture and module flow diagrams were drafted.

Core dependencies and libraries were installed and verified for compatibility.

Week 3–4 (September 18 – September 30): ARP scanner module was developed for detecting active devices. Initial dashboard UI with tables and basic charts was designed. Backend connectivity tests ensured successful communication between UI and Flask APIs. Improved JSON response formatting and field parsing for device data. Basic session handling and navigation structure of the dashboard were implemented.

Week 5–6 (October 01 – October 21): Dashboard modules were integrated with backend API routes. Live traffic monitoring and TShark-based packet capture were implemented and stabilized. Error-handling improvements and synchronization fixes were added for smoother real-time updates.

Bandwidth charts and activity timelines were linked to dynamic data sources. Basic access control and session timeout handling were added to improve security.

3. Final Outcomes :

By the end of Stage 1, NetVisor had evolved into a functional and stable prototype demonstrating the system's core capabilities. The multi-module architecture was successfully integrated, enabling seamless communication between the Flask backend, packet capture scripts, and the web dashboard. The system was able to: Detect all active devices on the network using ARP-based scanning Capture live packet data and extract essential fields such as protocol type, source/destination addresses, and timestamps Display real-time bandwidth usage through dynamic charts powered by Chart.js Visualize device activity and traffic patterns in an organized, user-friendly dashboard Maintain stable frontend–backend communication, ensuring accurate synchronization of monitoring modules Store captured logs and device information in MySQL for future analysis Support expansion for AI-driven anomaly detection, which is planned for Stage The system demonstrated smooth performance under testing conditions and proved capable of maintaining continuous monitoring without data loss or crashes. Although advanced features such as

References:

- [1] A. H. Lashkari, G. D. Gil, M. Mamun, and A. A. Ghorbani, “Characterization of VPN Traffic Using Time-related Features,” Proceedings of the 2nd International Conference on Information Systems Security and Privacy (ICISSP), pp. 407–414, 2016, doi: 10.5220/0005668404070414.
- [2] A. Aceto, D. Ciuonzo, A. Montieri, and A. Pescapè, “Mobile Encrypted Traffic Classification Using Deep Learning: Experimental Evaluation, Lessons Learned, and Challenges,” IEEE Transactions on Network and Service Management, vol. 16, no. 2, pp. 445–458, 2019, doi: 10.1109/TNSM.2019.2899082.
- [3] S. Rezaei and X. Liu, “Deep Learning for Encrypted Traffic Classification: An Overview,” IEEE Communications Magazine, vol. 57, no. 5, pp. 76–81, 2019, doi: 10.1109/MCOM.2019.1800770.
- [4] M. Conti, A. Dehghantanha, K. Franke, and S. Watson, “Internet of Things Security and Forensics: Challenges and Opportunities,” Future Generation Computer Systems, vol. 78, pp. 544546, 2018, doi: 10.1016/j.future.2017.07.052.
- [5] K. S. Kim and J. Y. Bang, “Real-Time Network Traffic Analysis and Anomaly Detection Using Python and Machine Learning,” IEEE Access, vol. 9, pp. 61250–61260, 2021, doi: 10.1109/ACCESS.2021.3074602.
- [6] Wireshark Foundation, “Wireshark User Guide – Packet Capture and Analysis,” Available: <https://www.wireshark.org/docs/>.

Guide

Project Coordinators

HOD-ISE

Prof. Vinutha Raghu
(Signature with Date)

Dr. Abdul Khadar A
(Signature with Date)

Dr. Pankaj Kumar
(Signatures with Date)

Prof. Mithuna HR
(Signatures with Date)